$$d_t = \|p_t^{\text{tar}} - p_t^e\|$$

$$r_t^G = \begin{cases} 1, & \text{target hit}, \\ \exp(-4d_t^2), & \text{otherwise}, \end{cases}$$

$$g_t = (p_t^{\text{tar}}, h_t), \qquad h_t \in \{0, 1\}$$